

Methodology & Data Cleaning

Panama Macroeconomic Data Cleaning Documentation

A Multi-Dataset ETL Pipeline for FDI, GDP, GVA & Provincial Indicators (2017–2023)

- Tools: Power BI, SQL, Excel
- Data Sources: INEC Panama (FDI, GDP, GVA, Provincial GDP, Quarterly GDP)

Project Overview

This project consolidates and cleans multiple macroeconomic datasets from INEC Panama to build a unified analytical foundation for studying. Foreign Direct Investment (FDI) Gross Value Added (GVA) Annual GDP Provincial GDP Quarterly GDP Sector-level economic performance

Standardized Data Cleaning Framework

All datasets were processed using the same senior-level ETL methodology:

Ingestion

Imported Excel/CSV files using SQL and Power BI.

Inspected raw structure to identify metadata, merged cells, footnotes, and non-data rows.

Structural Cleaning

Removed metadata rows, titles, footnotes, and totals. Example from the document:

- Removed irrelevant rows (1–9, 11, 30–34) that contained notes or were entirely empty.
- Promoted the correct header row.
- Renamed key identifier columns (e.g., Descripción → Sector, País de origen → Country).

Standardization

- Trimmed whitespace, corrected spelling inconsistencies, and harmonized sector/country names.
- Ensured consistent naming across datasets (e.g., “Sector”, “Year”, “GDP_Value”).

Reshaping

- Unpivoted all year columns into long format to support time-series analysis.

- Example from the document:

-Unpivoted the dataset from wide to long format for easier analysis.

Data Type Validation

- Converted Year → integer
- Converted GDP/FDI/GVA values → float
- leaved thousands separators and decimal inconsistencies.

Deduplication & Integrity Checks

Verified unique primary keys (Sector-Year, Country-Year, Province-Year).

Removed totals, subtotals, and alias rows.

Export

Saved each cleaned dataset as a separate CSV file with standardized naming conventions.

Dataset-Specific Cleaning Documentation

FDI by Economic Activity (2017–2023)

File: FDI_Economic_Activity_2017_2023.csv

Rows: 119 Columns: Economic Sector, Year, FDI_Value

Key Cleaning Actions

- Removed metadata rows and explanatory notes.
- Renamed first column to Economic Sector.
- Excluded provisional/inconsistent values (2020–2023).
- Removed participation % and YoY variation columns.
- Filled down merged sector names.
- Unpivoted 2017–2023 columns.

Document citation:

I removed the last two columns—participation percentage and year-over-year variation—since they weren't needed.

FDI by Country of Origin (Top 10, 2017–2023)

File: FDI_Country_Origin_2017_2023.csv

Rows: 70 Columns: Country, Year, FDI_Value

Key Cleaning Actions

- Removed metadata rows and empty rows.
- Dropped participation % and variation % columns.
- Removed national totals and “Otros países”.
- Standardized country names.
- Unpivoted year columns.
- Document citation:
- Excluded 'Otros países': Limited scope to only the top 10 countries

Gross Value Added (GVA) by Sector (2018–2022)

File: GVA_Sector_2018_2022.csv

Rows: 95 Columns: Sector, Year, GVA_Value

Key Cleaning Actions

- Removed metadata and non-data rows.
- Promoted correct headers.
- Standardized sector names.
- Removed subtotals and totals.
- Unpivoted year columns.
- Document citation:
- Removed subtotal and total rows... retaining only granular, sector-level data points.

GDP by Economic Activity (1996–2022)

File: GDP_Economic_Activity_1996_2022.csv

Rows: 433 Columns: Sector, Year, GDP_Value

Key Cleaning Actions

- Removed metadata and footnotes.
- Dropped non-data rows at top and bottom.

- Standardized sector descriptions.
- Unpivoted 1996–2022 columns.
- Validated numeric formats.
- Document citation:
- Removed Column 1 containing metadata and notes irrelevant to analysis.

Annual GDP (2018–2023)

File: GDP_Annual_2018_2023.csv

Rows: 139 Columns: Sector, Year, GDP_Value

Key Cleaning Actions

- Removed titles, footnotes, and totals.
- Standardized sector names.
- Unpivoted year columns.
- Cleaned numeric formatting.
- Document citation

Removed aggregated total rows and checked for duplicate sector-year entries.

Provincial GDP (2018–2023)

File: GDP_Provincial_2018_2023.csv

Rows: 60 Columns: Province, Year, P_GDP_Value

Key Cleaning Actions

- Removed national-level categories.
- Standardized province names.
- Unpivoted year columns.
- Removed duplicates and totals.
- Document citation
- Filtered sector rows... to focus only on GDP by province.

Quarterly GDP (2018–2023)

File: GDP_Quarterly_2018_2023.csv

Rows: 486 Columns: Sector, Quarter, Q_GDP_Value

Key Cleaning Actions

- Removed metadata and footnotes.
- Removed national totals (e.g., Valor Agregado Bruto).
- Standardized sector names.
- Unpivoted quarter columns.
- Cleaned numeric formatting.